

Model-based Design with Simulink



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R2026a

Agenda

- Why Model Based Design?
 - Simulink Overview
 - Introduction to Simulink Environment
 - Simulink Copilot
 - Find blocks, connect lines, run simulations and visualize output
 - Simulink Exercises
- Simulink Components
- File Management
- Agentic AI

Why Model Based Design?

What is Model-Based Design?

Systematic use of models throughout the development process

Modeling



Automation

Systematic use of **models** throughout the development process

Modeling

Simulation

Analysis



Automation



Systematic use of **models** throughout the development process

Modeling

Simulation

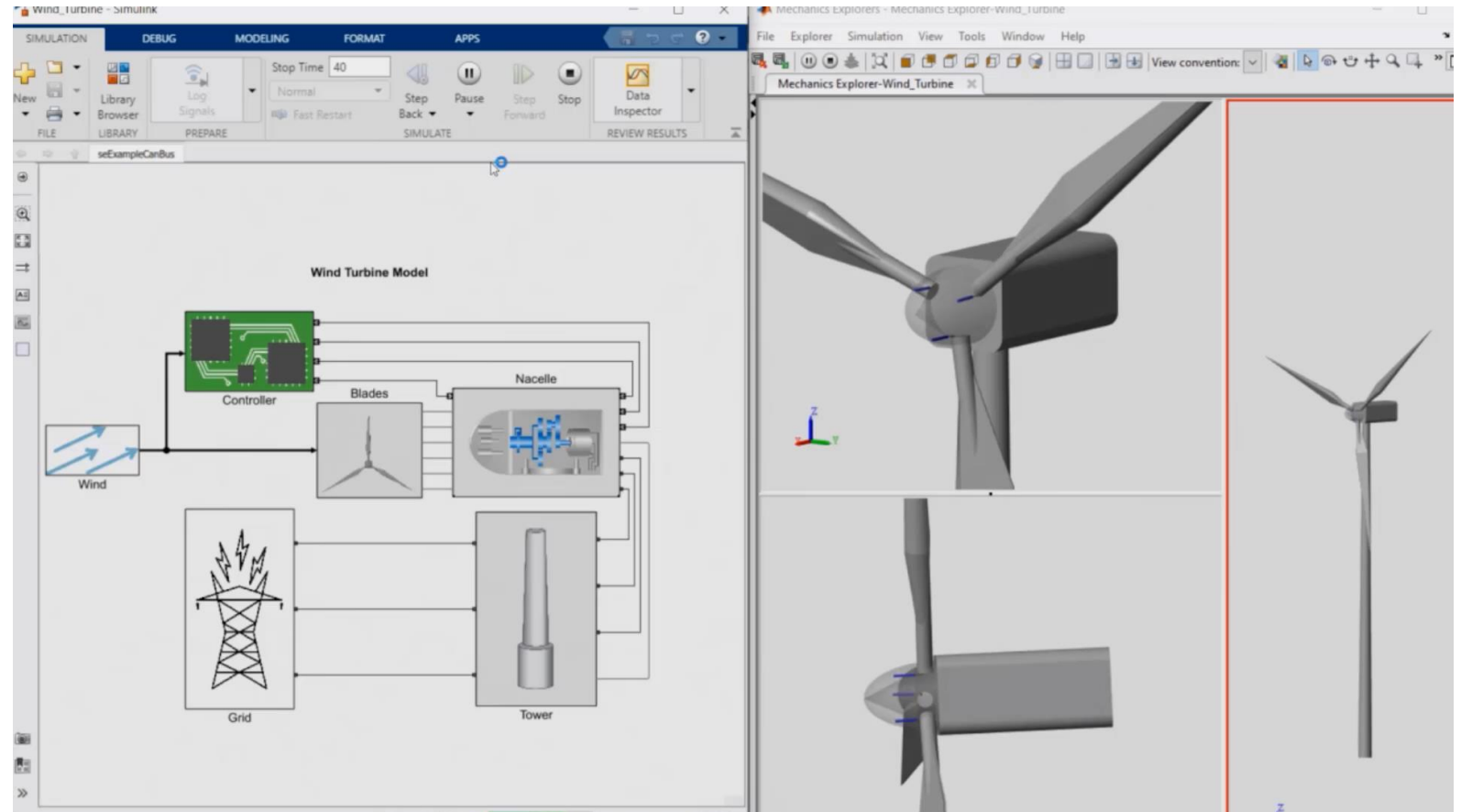
Analysis



Automation

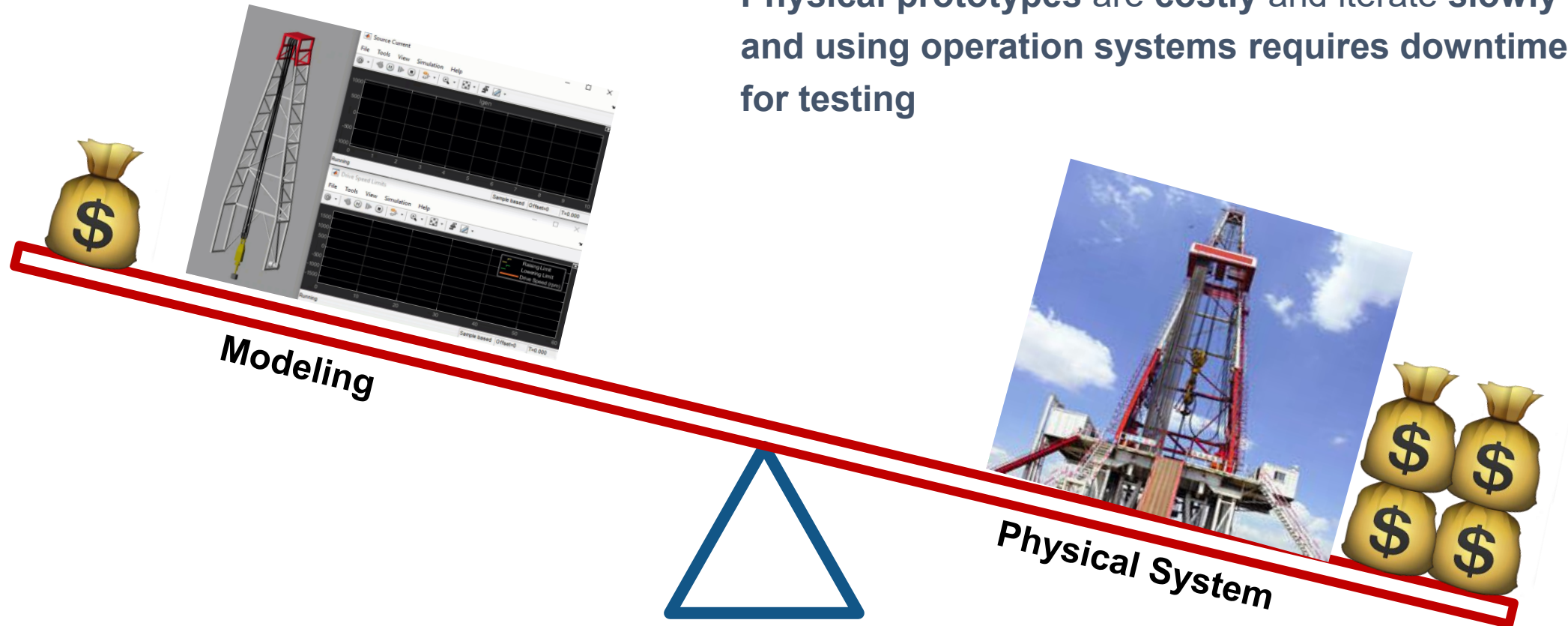
Coding

Verification



Traditional Development Process

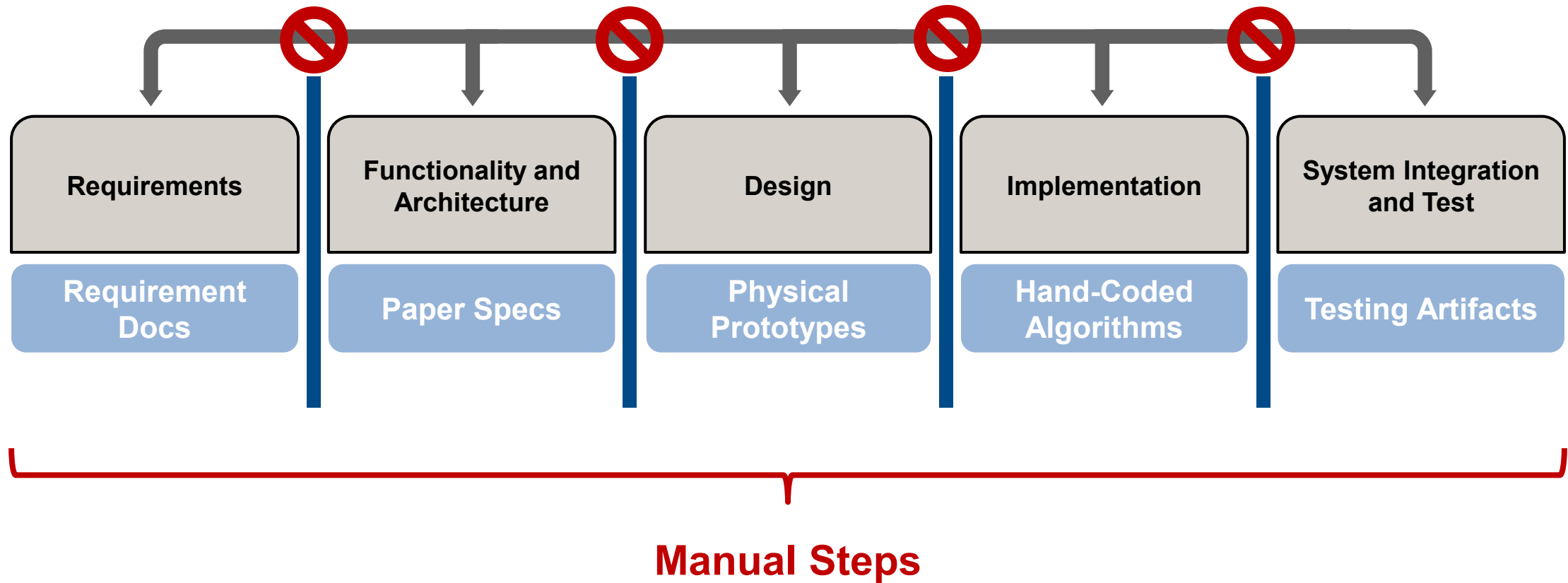
Physical prototypes are **costly** and iterate **slowly** and using operation systems requires **downtime** for testing



Design space **exploration**
Continuous design **improvement**
even after system is **operational**

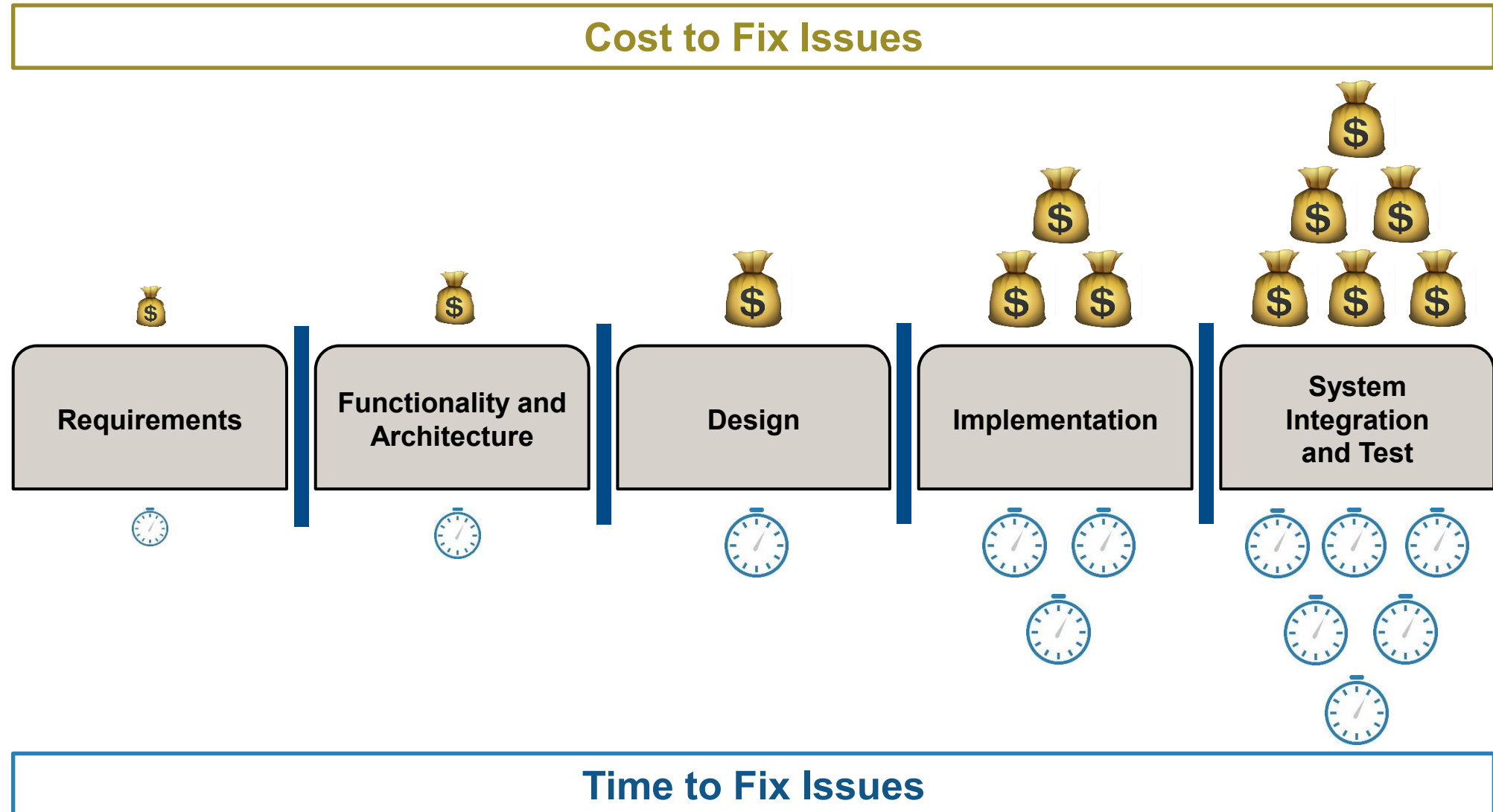
Costly and **time-consuming** to build
Hinders rapid iterations
Down-time and dangerous operation

Without Model-Based Design

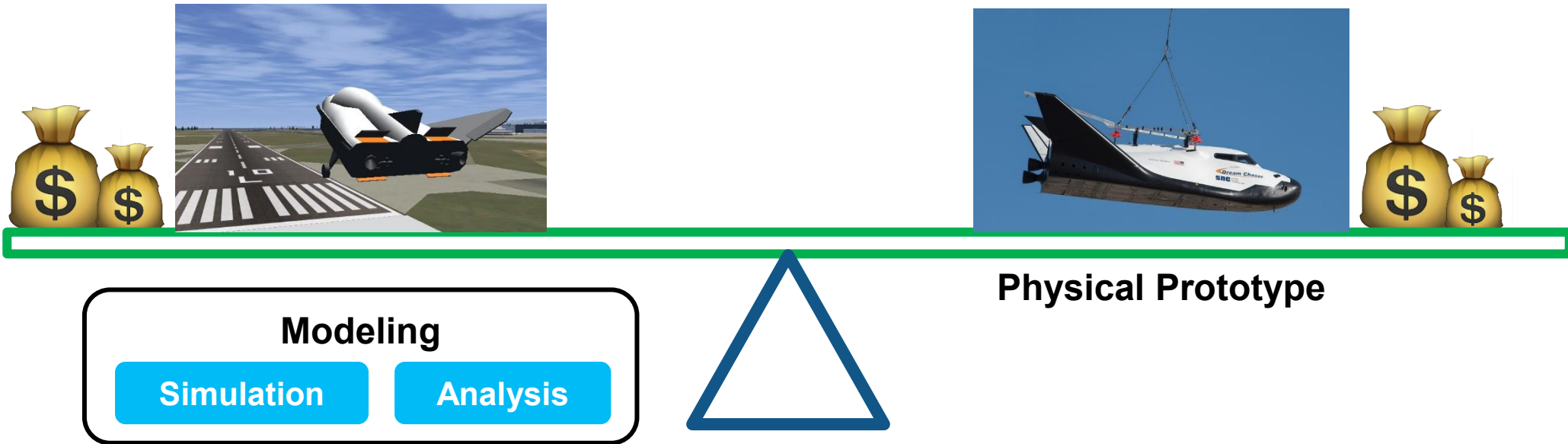


Manual steps introduce **errors** and **slow down** the development process

Issues found late in the process are more costly and time-consuming to fix

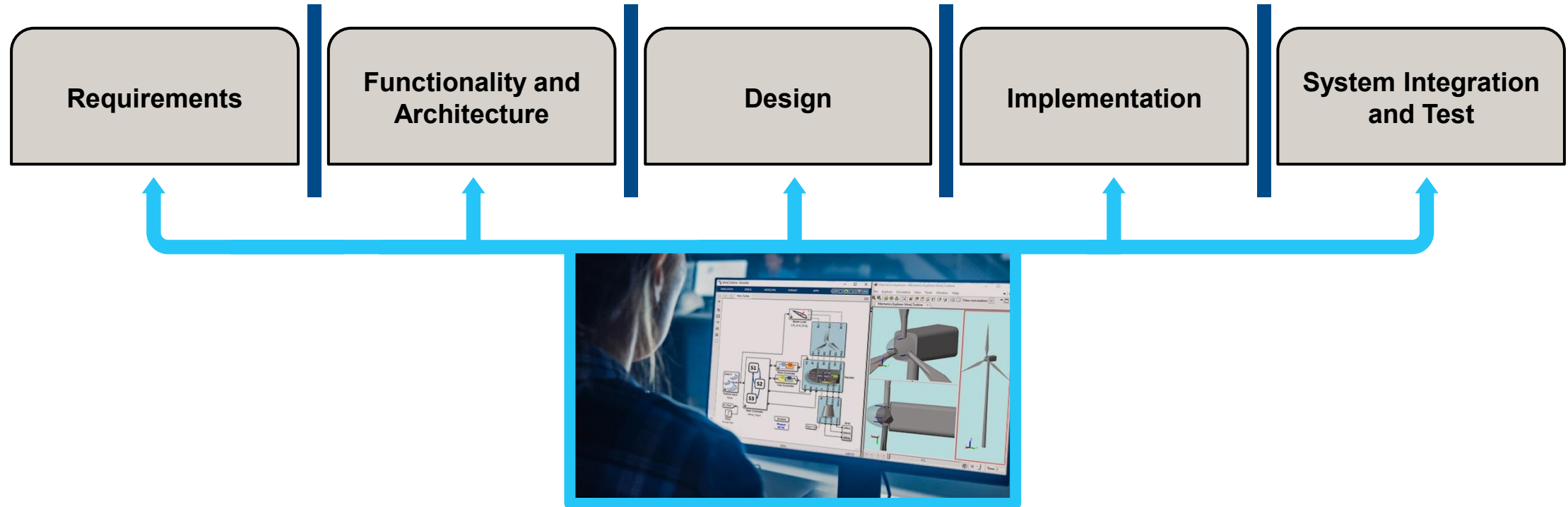


Modeling and simulation help cut costs and speed up design iterations



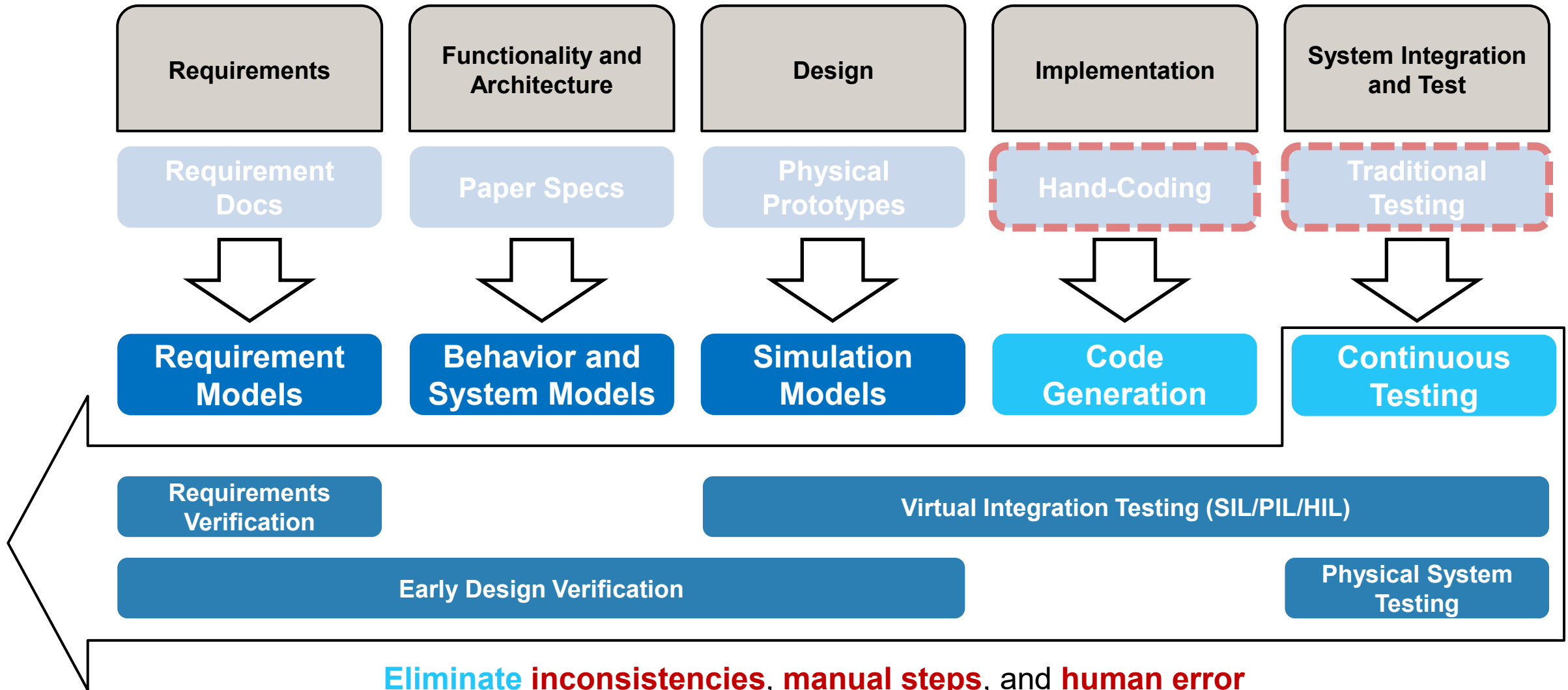
Rapid **prototyping** and
testing

With Model-Based Design



Models are at the **center** of your development process
Create a **digital thread**

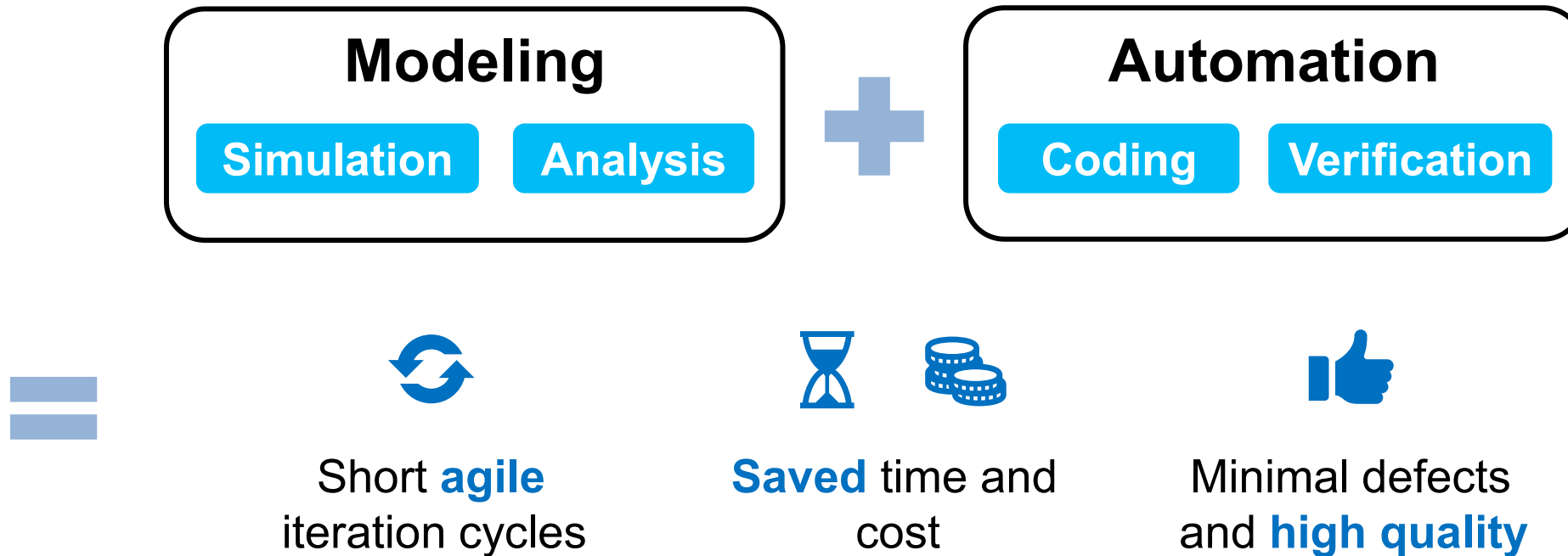
Automate steps in the design process with code generation and model-based verification



Eliminate inconsistencies, manual steps, and human error
So that you can **focus on** creating **innovative design** solutions

Model-Based Design

Systematic use of models throughout the development process



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- File Management
- Agentic AI

So then, how do we model?

Simulink

System modeling and simulation

Multi-domain modeling for many types of subsystems



MATLAB



SIMULINK

**Data-Driven
Models**

Machine learning
Deep learning

**Continuous-
Time Models**

System dynamics
Environment

**Discrete-Time
Models**

Digital control
DSP
Computer vision

**Physical
Models**

Electronics
Mechanics
Etc.

**State
Machines**

Control logic
Mode logic

**Discrete-
Event Models**

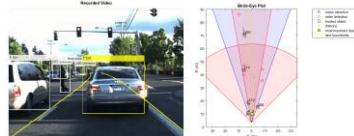
Servers/queues
Architecture

Analog/mixed-signal

Networks/ comms

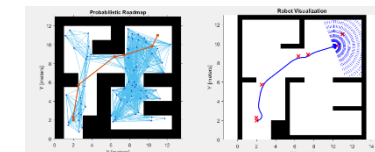
RF and radar

Systems of systems



Perception systems

Autonomous systems



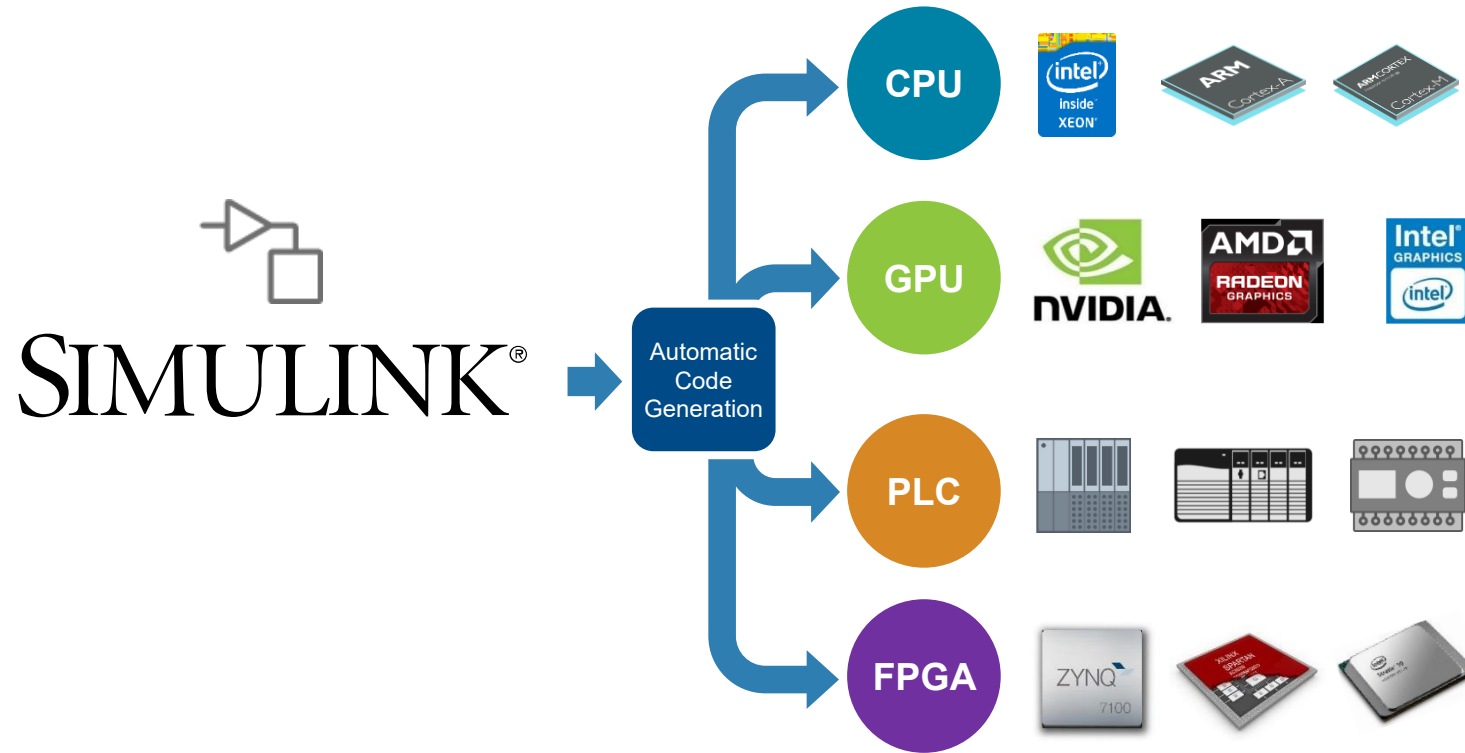
Simulink is the simulation **integration** platform

Ecosystem with **100+** third-party tools and languages



Automate **code generation** for prototyping and production

Reliable and **high performance**, with **flexible** choice of targets



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Accelerate your MATLAB and Simulink productivity with GenAI

AI Assistants



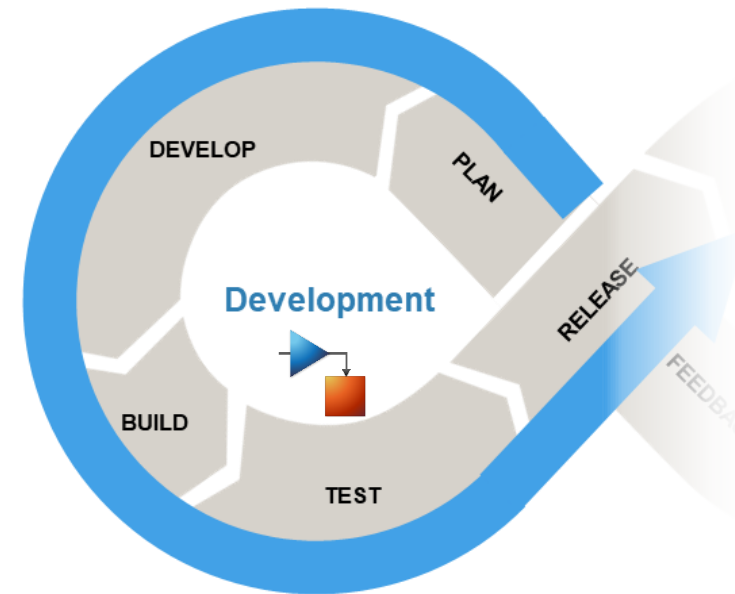
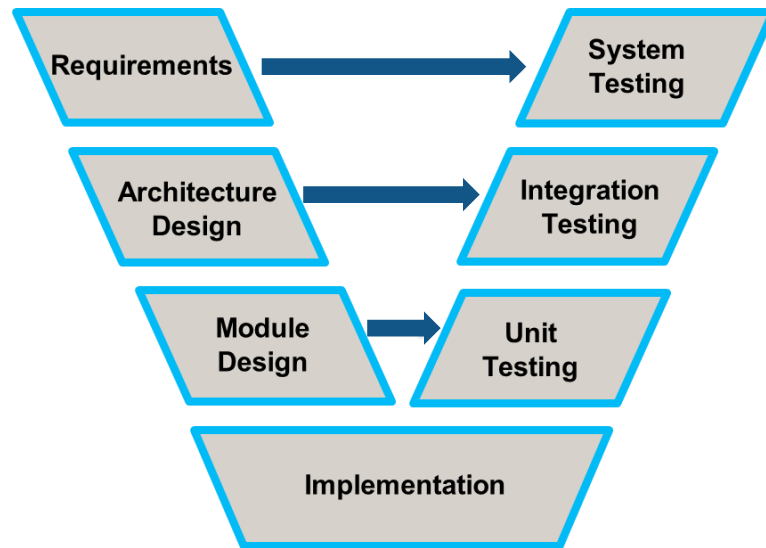
MATLAB and Simulink Copilots

optimized for MATLAB and Simulink

Simulink Copilot ✨

R2026a

An AI assistant optimized for Simulink and designed to improve your productivity with Model-Based Design



Simulink Copilot ✨

An AI assistant optimized for Simulink and designed to improve your productivity with Model-Based Design

Explain & Advise



- Learn tools & techniques
- Understand the model
- Search the model
- Get advice
- Troubleshoot errors

Review & Automate



- Automate tasks*
- Track progress*

* Grounded by your process
(through integration with Process Advisor)

R2026a Copilot is built to help with your everyday work

✓ Chat & Learn

The screenshot displays the Simulink environment for a 'BatteryManagementSystem' model. The main workspace shows a block diagram titled 'Battery Management System' with components including 'Scenario', 'Signal', 'BMS_Algorithms', 'BMS_Info', 'BMS_Cmd', 'BatteryPlantModel', 'KnownSOC', and 'Sensors'. The 'Copilot Chat' window is open on the right, featuring a header 'Simulink Copilot' and a welcome message: 'Hello! Simulink Copilot is here to assist you in understanding models, tools and errors.' Below this, a disclaimer states: 'Copilot sometimes produces output that seems accurate but is not. Validate generated output before use. Share feedback on the output to help improve the responses.' A button labeled 'Shuffle Example Prompts' is present. A list of example prompts includes: 'Help me understand this model in general.', 'What is happening in this model?', 'Explain the functionality and purpose of this model.', and 'Get familiar with Copilot's [limitations](#) before you start'. At the bottom of the chat window is a text input field with the placeholder 'Ask Simulink Copilot', a 'Deeper Insights' link, and a send button. The status bar at the bottom indicates 'Ready', '87%', and 'auto(ode3)'.

R2026a Copilot is built to help with your everyday work

- ✓ Chat & Learn
- ✓ Understand

The screenshot displays the Simulink R2026a software interface. The main workspace shows a Simulink model titled "Battery Management System". The model includes a "Scenario" block, a "BMS Algorithms" block, a "BatteryPlantModel" block, and a "Plant Model" block. The "BMS Algorithms" block contains sub-blocks for "BMS_Info", "BMS_Cmd", and "Sensors". The "BatteryPlantModel" block contains a "KnownSOC" block and "Sensors". The "Plant Model" block contains a "Sensors" block. The model is connected via "StateRequest" and "BMS_Cmd" signals.

The "Copilot Chat" window is open on the right side of the interface. It features a header "Simulink Copilot" and a greeting: "Hello! Simulink Copilot is here to assist you in understanding models, tools and errors." Below the greeting is a warning box: "Copilot sometimes produces output that seems accurate but is not. Validate generated output before use. Share feedback on the output to help improve the responses." A button labeled "Shuffle Example Prompts" is located below the warning box. A list of example prompts is provided:

- What are your capabilities?
- Give me a brief overview of control systems theory
- Tell me about this model
- Get familiar with Copilot's [limitations](#) before you start

At the bottom of the chat window is a text input field labeled "Ask Simulink Copilot" with a cursor. To the right of the input field is a "Deeper Insights" button and a "Send" button (represented by a right-pointing arrow). A footer note at the bottom of the chat window states: "Validate generated output before use."

The status bar at the bottom of the Simulink window shows "Ready", "87%", and "auto(ode3)".

R2026a Copilot is built to help with your everyday work

- ✓ Chat & Learn
- ✓ Understand
- ✓ Search

The screenshot displays the Simulink environment for a 'BatteryManagementSystem' model. The main workspace shows a block diagram titled 'Battery Management System' with components like 'Scenario', 'BMS_Algorithms', 'BMS_Info', 'BMS_Cmd', 'BatteryPlantModel', and 'Sensors'. The 'Copilot Chat' window on the right features a greeting from 'Simulink Copilot', a disclaimer about output accuracy, a 'Shuffle Example Prompts' button, and a list of suggested prompts: 'What are the outputs of this model?', 'Help me understand this model in general.', and 'What is happening in this model?'. It also includes a link to 'Get familiar with Copilot's limitations before you start' and an input field for asking questions.

BatteryManagementSystem - Simulink

SIMULATION DEBUG MODELING FORMAT APPS

Project New Open Save Print Library Browser Log Signals Add Viewer Signal Table Stop Time 5000 Normal Fast Restart Step Back Run Step Forward Stop Data Inspector Logic Analyzer Simulink Copilot Chat

BatteryManagementSystem

Battery Management System

Scenario Signal StateRequest BMS_Algorithms BMS_Info BMS_Cmd BatteryPlantModel KnownSOC KnownSOC Sensors

Copilot Chat

Simulink Copilot

Hello! Simulink Copilot is here to assist you in understanding models, tools and errors.

Copilot sometimes produces output that seems accurate but is not. Validate generated output before use. Share feedback on the output to help improve the responses.

Shuffle Example Prompts

- What are the outputs of this model?
- Help me understand this model in general.
- What is happening in this model?

Get familiar with Copilot's [limitations](#) before you start

Ask Simulink Copilot

Validate generated output before use.

Ready 87% auto(ode3)

R2026a Copilot is built to help with your everyday work

- ✓ Chat & Learn
- ✓ Understand
- ✓ Search
- ✓ Analyze & Implement

The screenshot displays the Simulink environment for a project named 'BatteryPlantModel'. The main workspace shows a block diagram titled 'Battery Plant Model'. This diagram includes a 'Pre Charge Circuit' block on the left, which outputs 'Vout_Charger' and 'Vout_Inverter' signals. These signals pass through 'Sensors' blocks and then through 'StateRequest' and 'BMS_Cmd' blocks. The final outputs are 'ChargerCircuitClk' and 'InverterCircuitClk', which are connected to a 'Charger and Load' block. The 'Copilot Chat' window is open on the right side of the interface. It features a header 'Simulink Copilot' and a greeting: 'Hello! Simulink Copilot is here to assist you in understanding models, tools and errors.' Below this, a warning box states: 'Copilot sometimes produces output that seems accurate but is not. Validate generated output before use. Share feedback on the output to help improve the responses.' A button labeled 'Shuffle Example Prompts' is visible. There are three example prompts listed: 'What engineering concept is implemented in this model?', 'Tell me about this model', and 'Tell me about the inputs and outputs of this model'. An information icon and text advise users to 'Get familiar with Copilot's [limitations](#) before you start'. At the bottom of the chat window is a text input field labeled 'Ask Simulink Copilot' with a 'Deeper Insights' button and a send icon. A status bar at the bottom of the Simulink window shows 'Ready', '80%', and 'FixedStepAuto'.

R2026a Get assistance where you need it

- ✓ Chat & Learn
- ✓ Understand
- ✓ Search
- ✓ Analyze & Implement
- ✓ Diagnose & Fix

The screenshot displays the Simulink environment for a 'BatteryPlantModel'. The main workspace shows a block diagram titled 'Battery Plant Model' with two primary components: 'Charger and Load' and 'Battery Pack'. The 'Charger and Load' block includes sub-models for 'StateRequest', 'BMSCmd', 'ChargerCircuitClc', and 'InverterCircuitClc'. It is connected to the 'Battery Pack' block via a 'Request' signal. The 'Battery Pack' block contains 'CellSOCs', 'CellVoltages', 'CellTemps', and 'PackVoltage' outputs. A 'Pack Current' signal is also shown. The interface includes a top toolbar with tabs for SIMULATION, DEBUG, MODELING, FORMAT, and APPS, along with various tool icons like 'Log Signals', 'Add Viewer', and 'Signal Table'. A 'Copilot Chat' window is open on the right, featuring a 'Simulink Copilot' header and a message: 'Hello! Simulink Copilot is here to assist you in understanding models, tools and errors.' Below this, a warning box states: 'Copilot sometimes produces output that seems accurate but is not. Validate generated output before use. Share feedback on the output to help improve the responses.' A 'Shuffle Example Prompts' button is visible. A list of example prompts includes: 'What engineering concept is implemented in this model?', 'Tell me about this model', 'Tell me about the inputs and outputs of this model', and 'Get familiar with Copilot's limitations before you start'. At the bottom of the chat window is an input field labeled 'Ask Simulink Copilot' and a 'Deeper Insights' link. The status bar at the bottom indicates 'Ready', 'View 2 errors', '100%', and 'FixedStepAuto'.

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- Simulink Components
- File Management
- Agentic AI

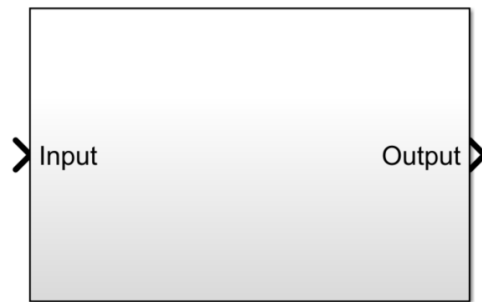
Let's give it a try

Agenda

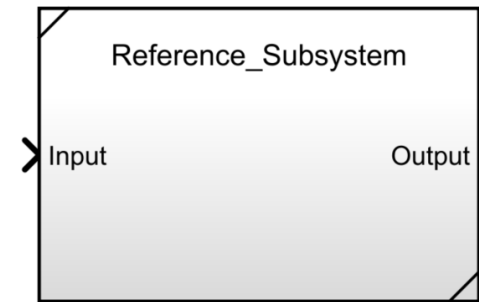
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What is a Simulink Component?

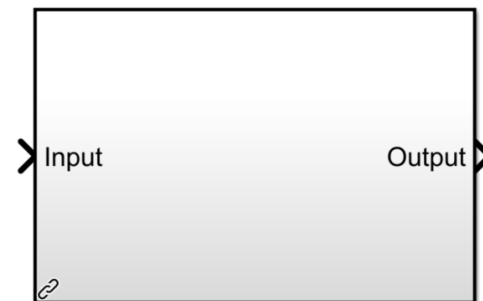
- In the context of Simulink modeling, a component is: A piece of your design that has an interface and facilitates assembly of a system or a larger component.



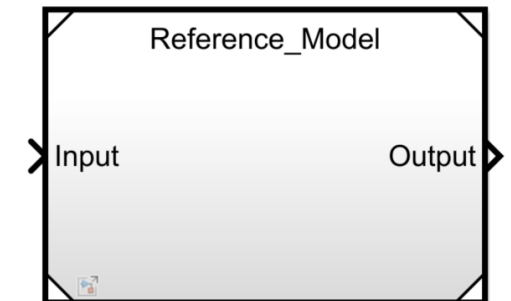
In-model Subsystem



Subsystem Reference

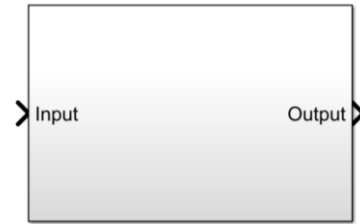


Linked Subsystem



Model Reference

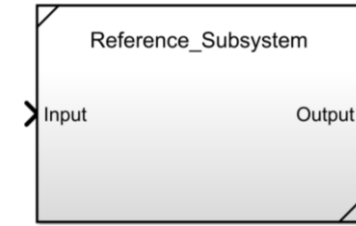
Architecture



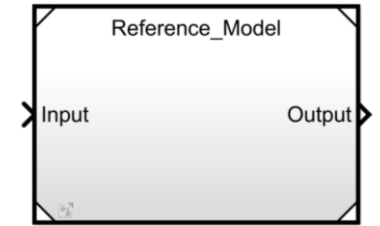
In-model Subsystem



Linked Subsystem



Subsystem Reference



Model Reference

Storage	With parent model	Multiple Linked subsystems per file (library)	Separate file	Separate file
Boundary type	Virtual or Atomic	Virtual or Atomic	Virtual or Atomic	Always atomic
Incremental loading	Not supported	Supported	Supported	Supported
Reduced memory usage when there are multiple instances	Not supported	Not supported	Not supported	Supported in Accelerator and Rapid Accelerator modes
Standalone behavior	Not supported	Not supported	Not supported	Supported
Simulink cache	Not supported	Not supported	Not supported	Supported
Physical connections at interface	Supported	Supported	Supported	Not supported
Design for explicit code reuse	No	Yes	Yes	Yes

Selecting the right component for each use case

If you want to...	Use..
Visually organize a group of blocks or components	In-Model Virtual Subsystem
Encapsulate small pieces of code in non-reusable packages	In-Model Atomic Subsystem
Create utilities that are widely reused and change infrequently	Linked Subsystems (Libraries)
Create a reusable component that supports physical connection (Simscape) interfaces	(Virtual) Subsystem Reference
Develop/simulate a component as a standalone model	Model Reference
Create a version of your component with IP protection	Model Reference
Use accelerator mode to speed up subsequent instances/runs of a component	Model Reference

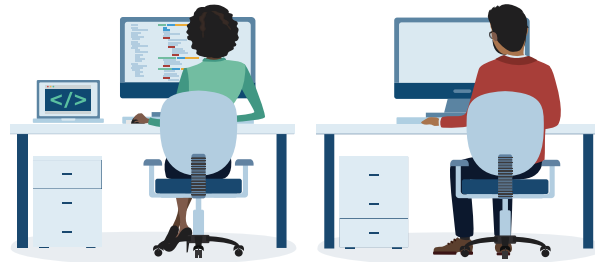
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Why set up Projects?

Do any of these sound familiar?

“It works on my computer,
but not on yours...”



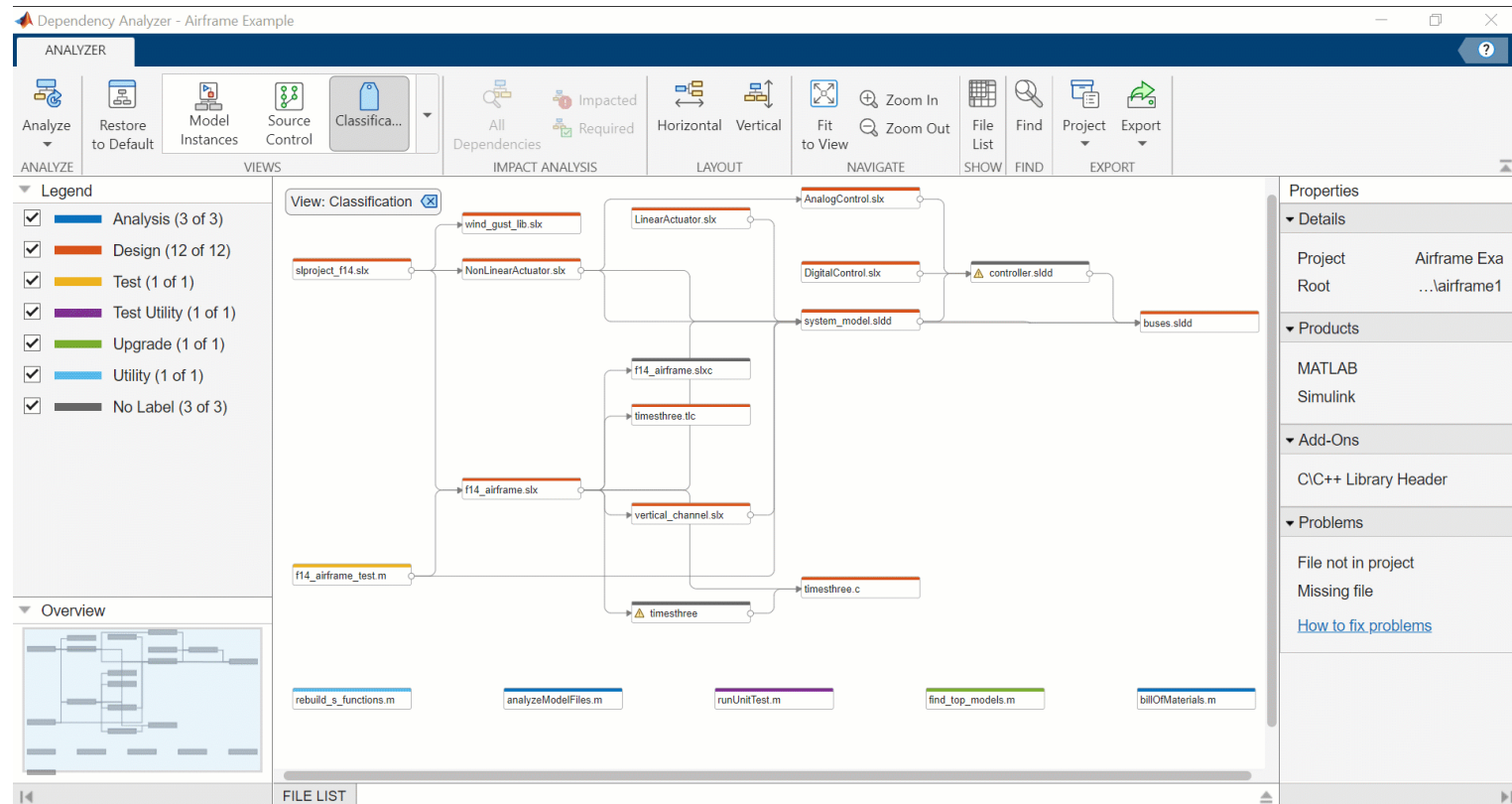
Ever get instructions like this?

How to simulate and test my work

1. Download Zip File
2. Unzip into C:\
3. Open MATLAB
4. Clear Preferences
5. Add “Data” folder to path
6. Remove “Network\CommonMATLAB” from Path
7. Run the script “starting_stuff.m”
8. Run these 3 lines of code
 - a. `set_param('model_config','Time','10')`
 - b. `set_param('model_config','TimeStep','0.01')`
 - c. `set_param('model_config','tlc','ert')`
9. Open the model “myCoolModel.slx”
10. Hit Ctrl+D to make sure it updates
11. Now do the boogie woogie
12. Open the model “myCoolestModel.slx”
13. Clear the variable ‘xGains’ from base workspace
14. Simulate the model
15. Setup mex for C code generation by running
 - a. `mex -setup C`
16. Close the model “myCoolModel.slx”
17. Now turn yourself around
18. `savepath`
19. Exit MATLAB
20. And that's what its all about!

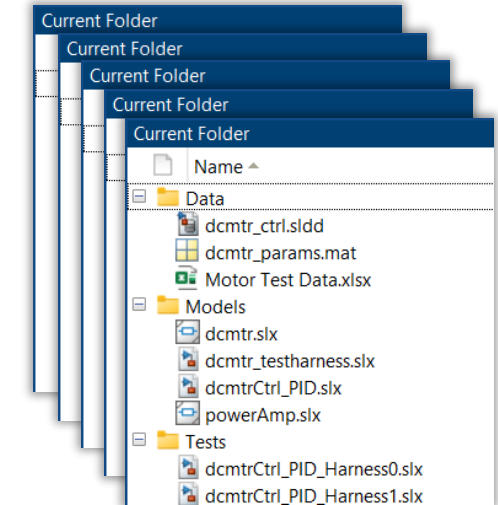
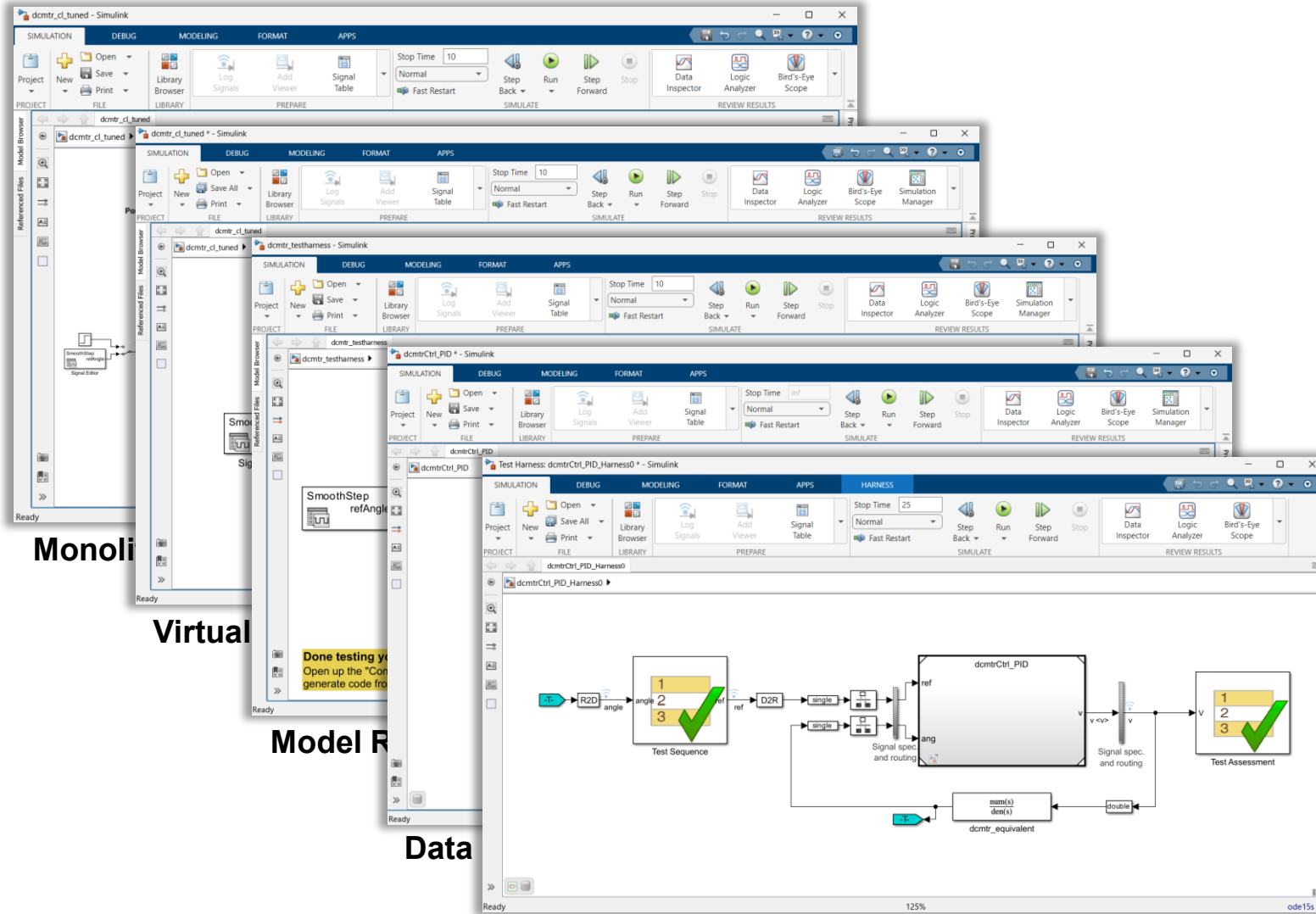
Dependency Analyzer

- Dependency Analyzer can:
 - Find required products
 - Find dependencies among project files
 - and fix problems
 - Run impact analysis to qualify changes



Why set up Projects?

The progression of a Simulink project...

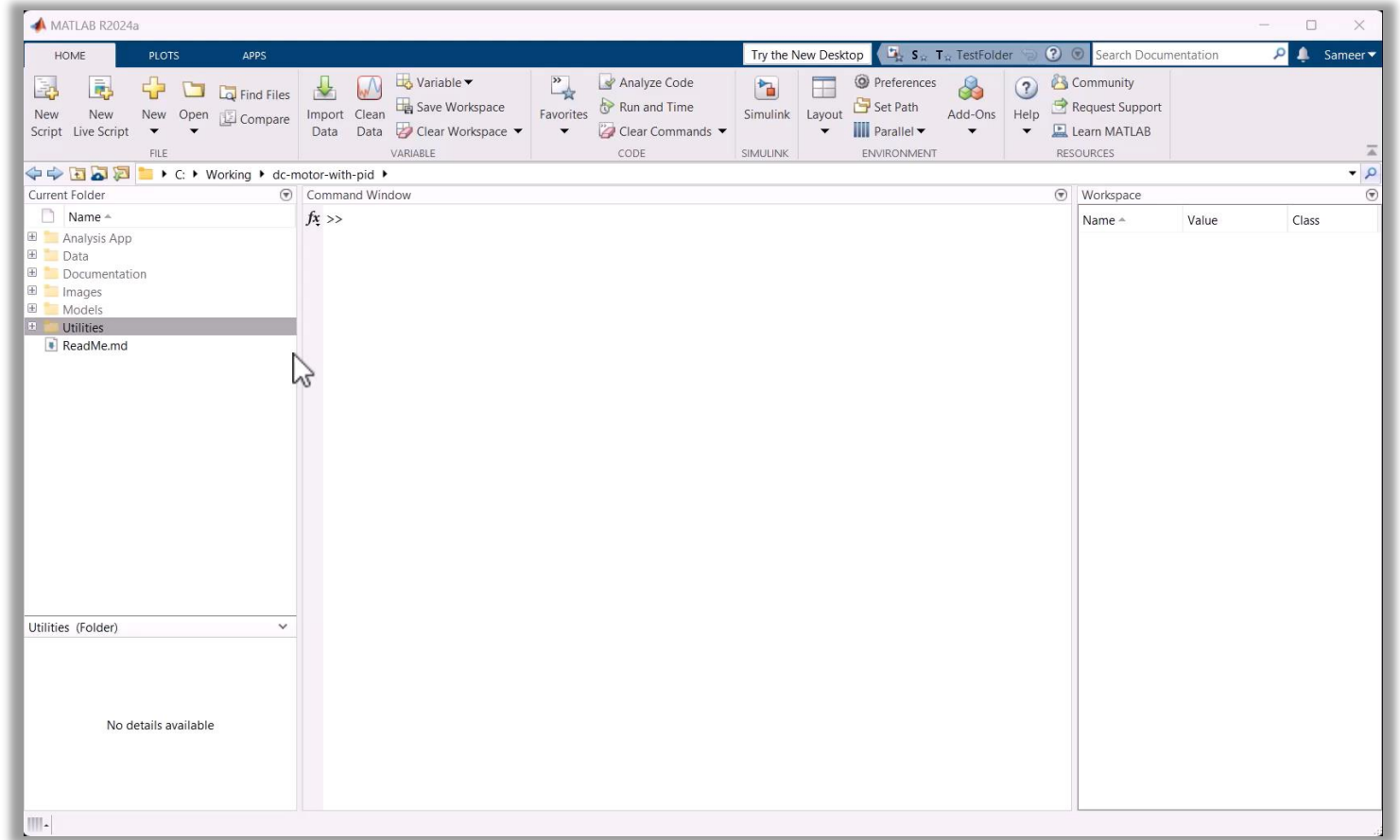


How do you manage all your files & folders?

Easy Environment Setup

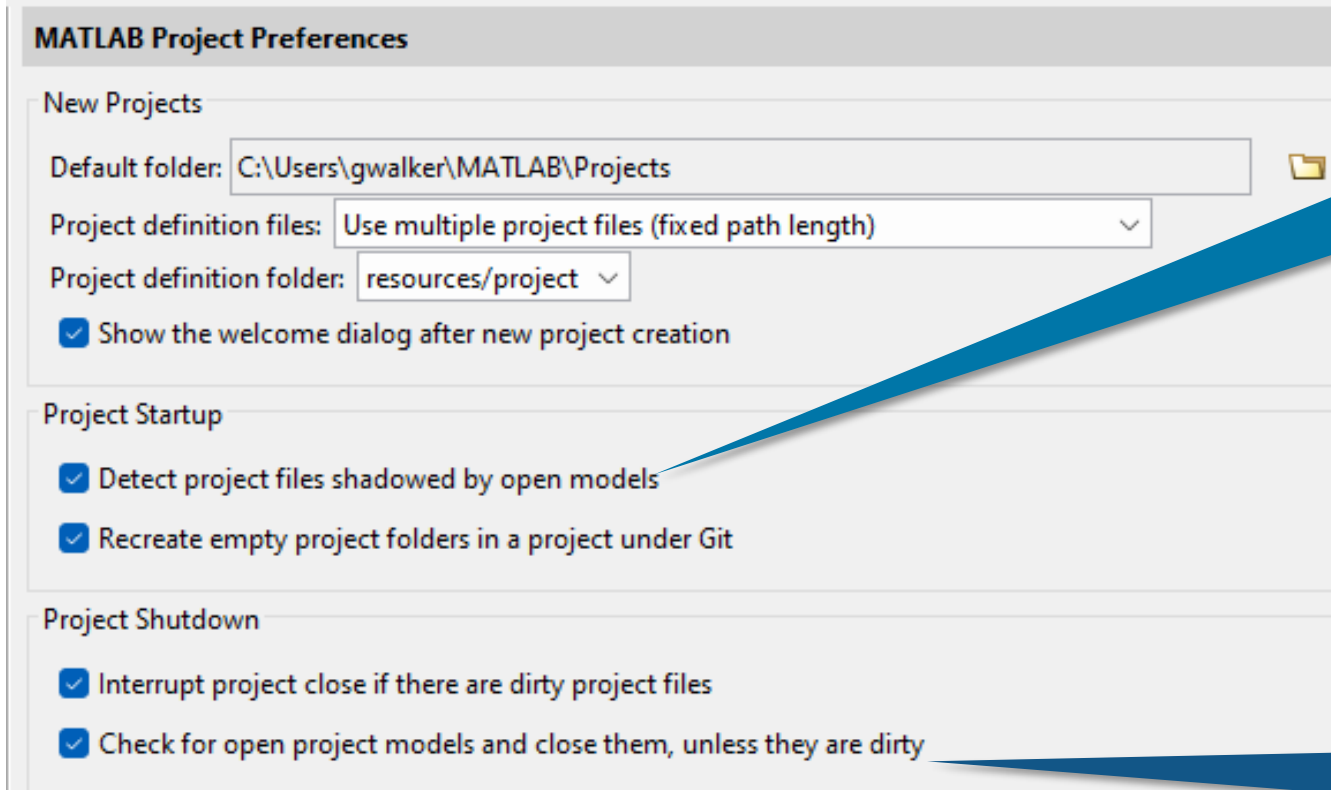
Getting started with Projects

- Setup Path
- Configure startup/shutdown functions
- Create shortcuts for common tasks
- Manage generated artifacts



Easy Environment Setup

“Mistake-proofing” your work through projects



The image shows the MATLAB Project Preferences dialog box. It is divided into three sections: New Projects, Project Startup, and Project Shutdown. In the New Projects section, the Default folder is set to C:\Users\gwalker\MATLAB\Projects, Project definition files are set to Use multiple project files (fixed path length), and Project definition folder is set to resources/project. In the Project Startup section, both Detect project files shadowed by open models and Recreate empty project folders in a project under Git are checked. In the Project Shutdown section, both Interrupt project close if there are dirty project files and Check for open project models and close them, unless they are dirty are checked. A blue arrow points from the 'Detect project files shadowed by open models' checkbox to a blue callout box on the right.

MATLAB Project Preferences

New Projects

Default folder: C:\Users\gwalker\MATLAB\Projects

Project definition files: Use multiple project files (fixed path length)

Project definition folder: resources/project

☒ Show the welcome dialog after new project creation

Project Startup

☒ Detect project files shadowed by open models

☒ Recreate empty project folders in a project under Git

Project Shutdown

☒ Interrupt project close if there are dirty project files

☒ Check for open project models and close them, unless they are dirty

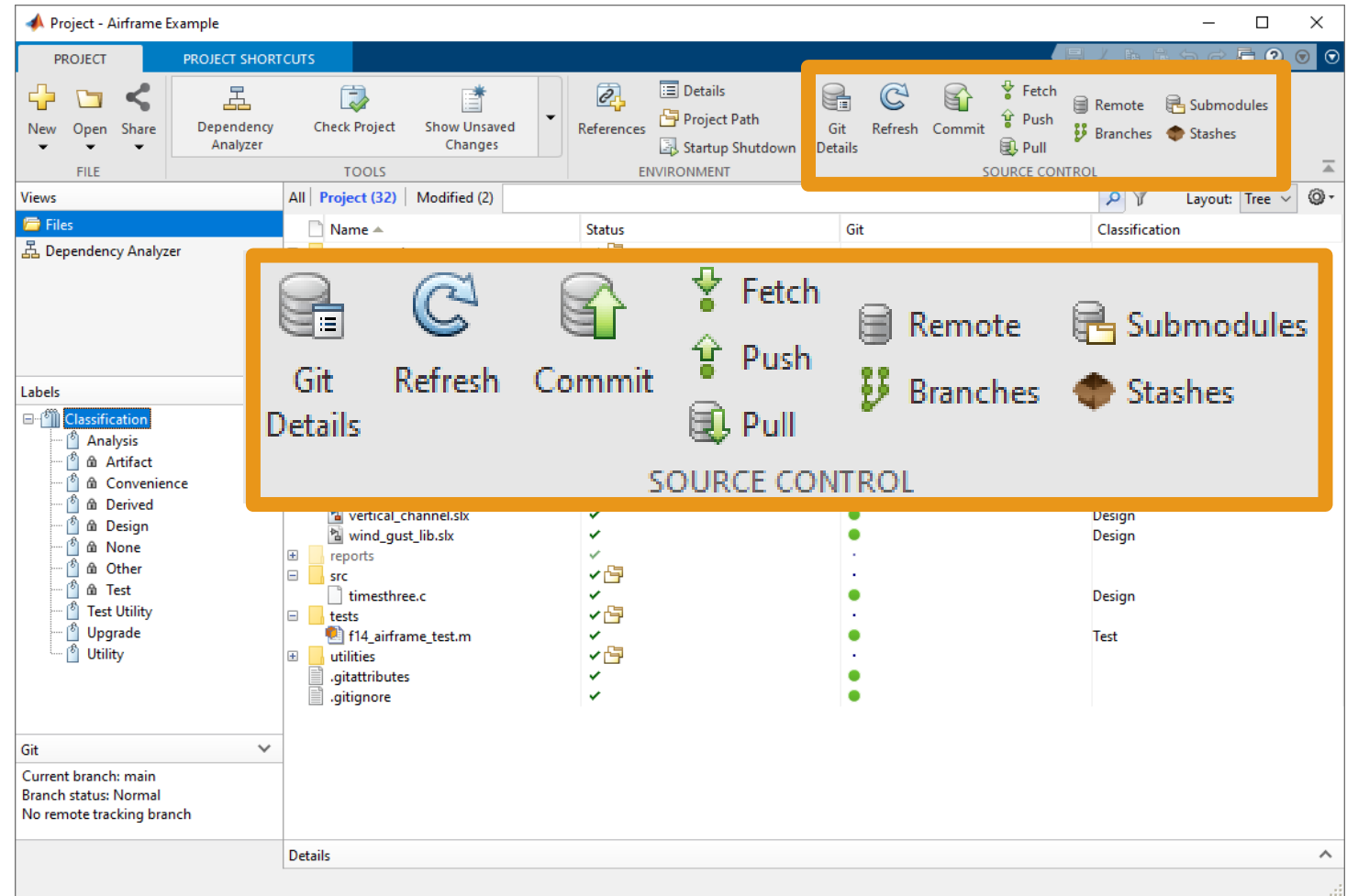
Shadowed models & libraries can give unexpected results: the project finds them and warns you

Unsaved changes risk losing work or getting confusing results: the project will ask you to confirm

Managing source control with projects

Built-in support for Git:

- Refresh
- Pull
- Commit
- Push
- Branches (history)



[Use Git with MATLAB](#)

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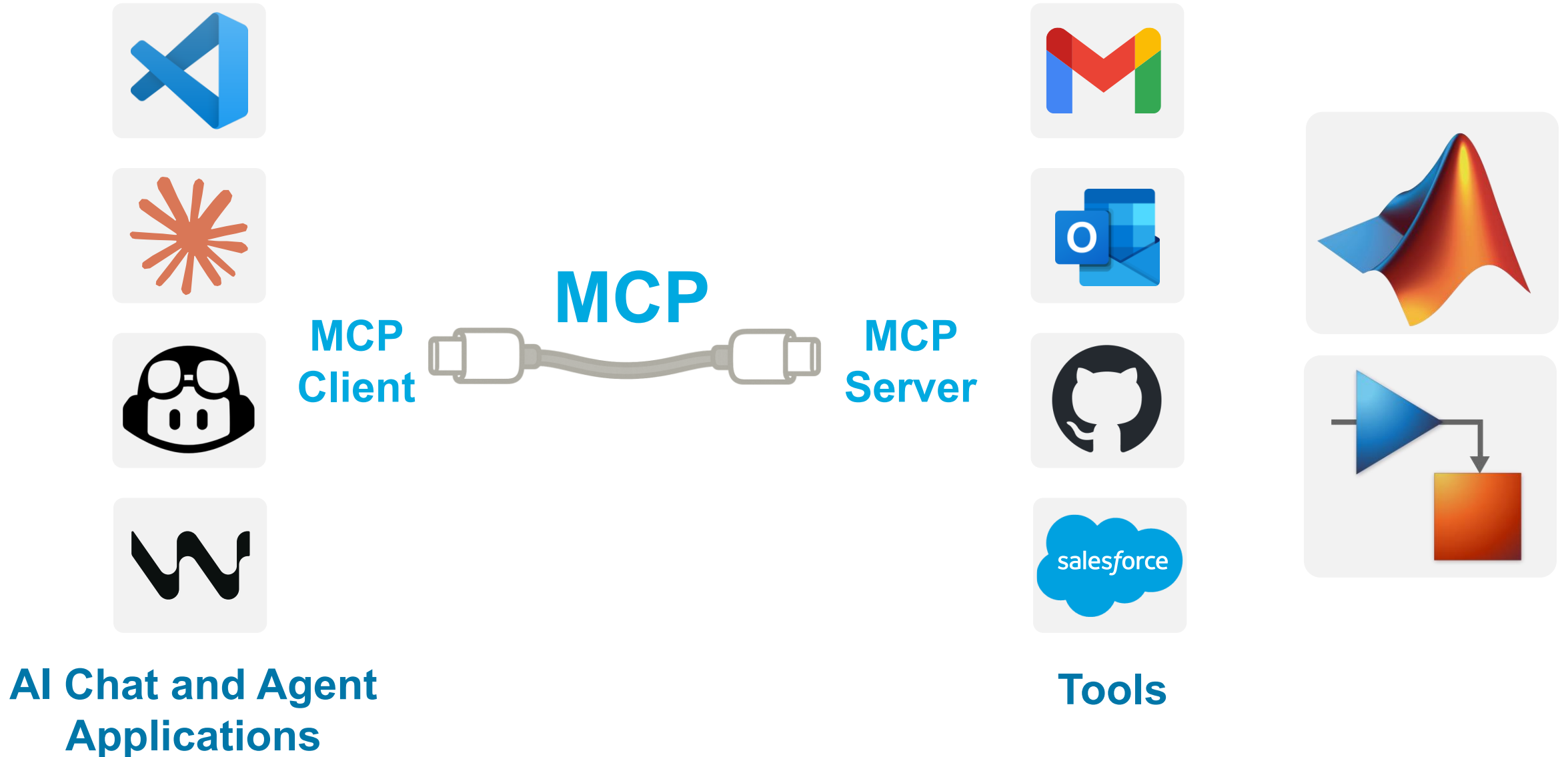
Agentic Workflows



External GenAI Agents

*use MATLAB or Simulink with external Agents
via MCP or VS Code*

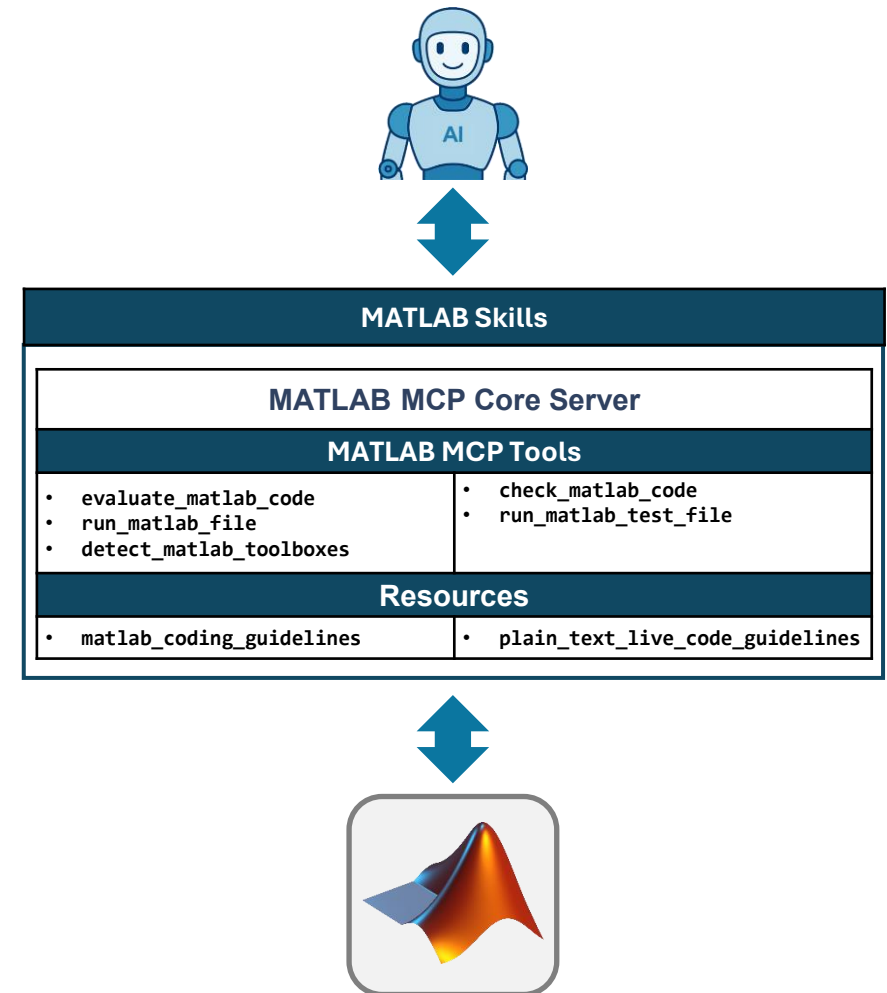
Agentic workflows are enabled by Model Context Protocol (MCP)



MATLAB Agentic Toolkit

Connect trusted MATLAB capabilities to AI agents to make engineering and scientific workflows agent-ready

- A **free, open-source** local MCP server that connects GenAI assistants to a **locally licensed** MATLAB, enabling intelligent automation and code execution
- Expert Skills Catalog encode MATLAB best practices, guiding AI in testing, debugging, and code modernization.

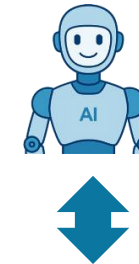


<https://github.com/matlab/matlab-agentic-toolkit>

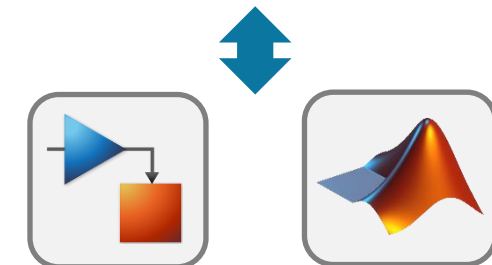
Simulink Agentic Toolkit

Connect agents to Simulink giving them the ability and the knowledge to work with Simulink models effectively.

- Builds on top of the MATLAB MCP Core Server
- MCP tools for reading, editing, querying, testing, and checking Simulink models
- Agent skills encoding Model Based Design best practices for model building, plant specification, testing, requirements, and tool initialization



Simulink Skills	
Simulink MCP Tools	
- model_overview - model_read - model_query_params - model_resolve_params	- model_edit - model_test - model_check
MATLAB MCP Core Server	
MATLAB MCP Tools	
Resources	



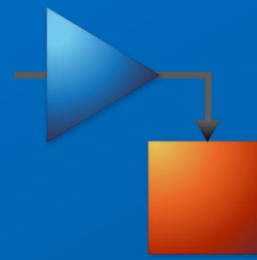
<https://github.com/matlab/simulink-agentic-toolkit>

Demo

Feedback



<https://share-na2.hsforms.com/1zTteBBGuTDC3z7UtwBNbHw3sou6>

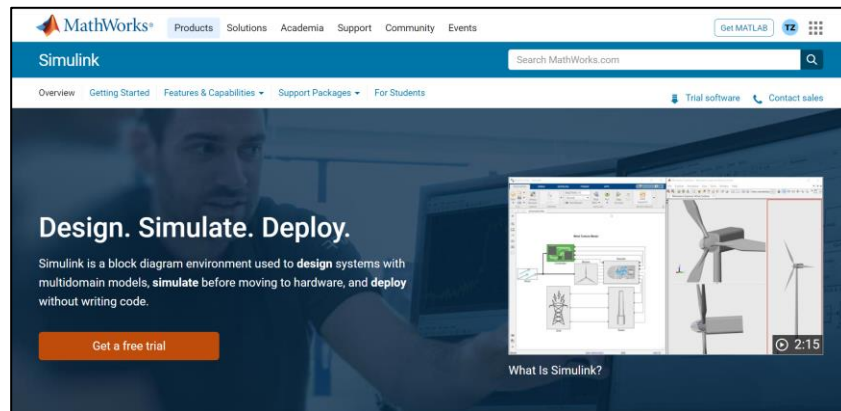


SIMULINK®

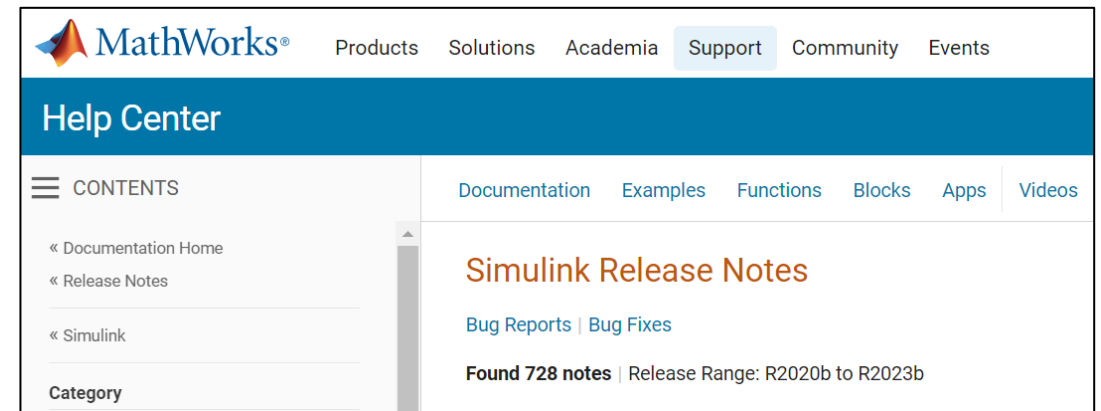
Q&A session

What you should check out

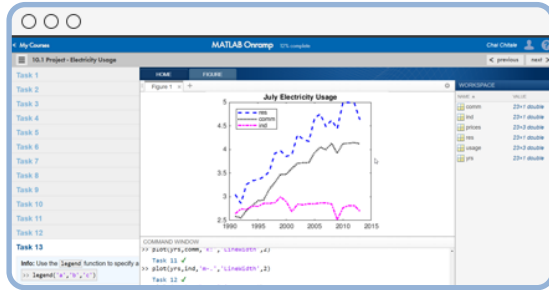
[Simulink Product Page](#)



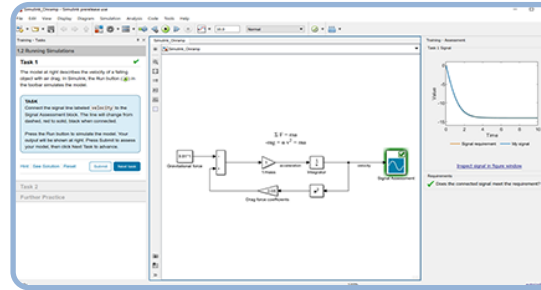
[Release Notes](#)



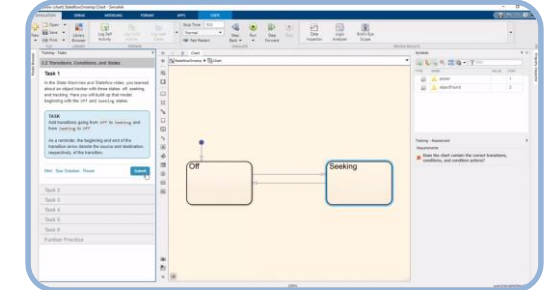
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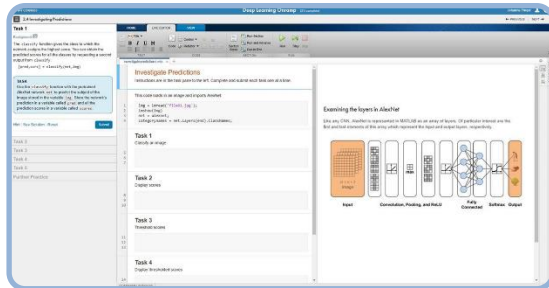
MATLAB Onramp



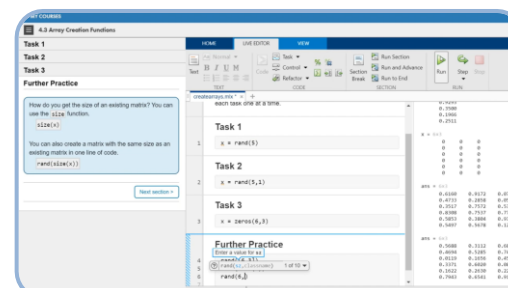
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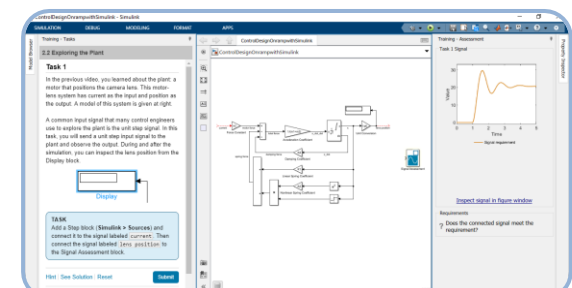
Stateflow Onramp



Deep Learning Onramp



Machine Learning Onramp



Control Design Onramp